

But these efforts can only go so far if the products themselves are deficient. Before the arrival of the market index fund in asset management, investors had no cheap way into a broad, diversified stock portfolio. They were stuck with expensive active management products, and active management on average underperforms the market (see chapter 16). The market index fund changed everything. Its introduction took immense effort by dedicated individuals, a large dollop of investor education, and the firms that pioneered it were relative outsiders to the asset management industry. And it almost didn't happen—the first attempt failed. Regime shifts are rarely engineered by companies profiting from the status quo.⁶⁶

The annuity market is similarly ripe for innovation. There is currently no annuity product similar to the market index fund: an annuity that is dirt cheap, simple, can be sold in scale, avoids costly intermediaries, and is actually bought by investors rather than sold to them. There are more challenges here compared to the market index fund because insurance regulation varies by state, and adverse selection makes pricing tricky. But what a benefit this would be to investors!

4.5. ARE WE DIS-SAVING ENOUGH IN RETIREMENT?

For those lucky, or prescient enough, to enjoy a comfortable retirement, there is a final *retirement savings puzzle*: the elderly tend to dis-save much more slowly than predicted by simple life-cycle models. In fact, for those with relatively large retirement savings, Poterba, Venti, and Wise (2011b) show that retirement account balances continue to increase after individuals leave the workforce! Fewer than 20% of households withdraw prior to age 70.5, the age when households are forced by law to take minimum distributions, and when they do withdraw, the withdrawal rate is low. Even after age 70.5, the withdrawal rate tends to be below the rate of return earned on retirement assets. Not surprisingly, many people die with piles of savings.

The need to pay for medical care is one big reason. Another is the attachment to one's house, which is the largest investment for most individuals.⁶⁷ (Houses are highly illiquid assets with huge idiosyncratic risk, funded by a short position in bonds.) While the basic life-cycle model predicts that agents should reduce their housing when they retire, homeowners choose to stay in their homes for as long as possible and cannot easily borrow against them. This can be alleviated by reverse mortgages, but this market is still small and suffers from many of the same drawbacks as the annuity market. Housing wealth is not typically used to support

⁶⁶ See Christensen (1997).

⁶⁷ See Venti and Wise (2002). Nakajima and Telyukova (2012) show this behavior results from including housing in a life cycle model. Fischer and Stamos (2013) also develop a life-cycle model with housing.